

LIVERMORE, CA, USA

WMO: 724927

Lat: 37.693N

Lon: 121.814W

Elev: 120

StdP: 99.89

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23285

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	0.8	-8.8	1.8	11.0	-6.3	2.2	11.1	8.9	10.6	7.9	11.8	0.7	70	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	37.2	19.7	35.0	18.8	32.9	18.3	20.8	34.0	19.9	32.4	19.0	30.8	4.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.6	24.4	15.1	10.9	22.5	14.3	10.3	21.4	60.3	34.1	56.9	32.3	54.2	31.1	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.9	7.2	DB	-2.8	40.9	1.1	1.6	-3.6	42.0	-4.3	42.9	-4.9	43.8	-5.7	45.0	(4)
			WB	-3.9	22.3	1.2	1.5	-4.8	23.5	-5.5	24.4	-6.2	25.2	-7.1	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	9.1	10.4	12.4	14.1	16.9	20.3	22.5	22.4	21.3	17.6	12.4	9.1	(6)
(7)		DBStd	5.79	2.71	2.70	2.90	3.17	3.27	3.56	3.25	2.79	3.19	2.97	3.05	2.82	(7)
(8)		HDD10.0	151	51	26	10	2	0	0	0	0	0	12	51		(8)
(9)		HDD18.3	1437	287	222	186	133	68	16	3	2	8	49	179	286	(9)
(10)		CDD10.0	2247	22	37	83	126	215	310	388	384	340	237	83	23	(10)
(11)		CDD18.3	491	0	0	1	6	24	76	132	128	98	27	1	0	(11)
(12)		CDH23.3	6061	0	1	32	149	412	1015	1502	1362	1143	423	22	0	(12)
(13)		CDH26.7	2938	0	0	2	44	176	512	784	687	584	148	1	0	(13)
(14)	Wind	WSAvg	2.8	2.1	2.2	2.4	3.0	3.6	3.8	3.9	3.5	2.8	2.2	1.8	2.1	(14)
(15)	Precipitation	PrecAvg	378	73	65	57	27	10	2	1	2	5	20	50	62	(15)
(16)		PrecMax	822	169	185	169	118	51	18	18	23	38	92	138	178	(16)
(17)		PrecMin	162	8	2	2	4	0	0	0	0	0	0	2		(17)
(18)		PrecStd	132	52	49	43	25	13	4	3	4	9	22	40	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	22.7	26.4	31.1	35.0	38.9	39.8	38.9	38.1	33.7	25.9	19.5	(19)
(20)			MCWB	11.1	13.5	15.8	17.1	18.8	20.5	20.7	20.1	19.1	17.6	14.7	12.7	(20)
(21)		2%	DB	17.9	20.3	23.9	27.5	31.3	35.4	36.5	36.0	35.7	30.3	23.3	17.4	(21)
(22)			MCWB	11.2	12.9	14.8	16.4	17.5	19.2	19.8	19.1	18.5	16.1	14.2	11.7	(22)
(23)		5%	DB	16.1	18.0	21.4	24.3	28.0	32.6	33.9	33.2	33.0	27.9	20.6	16.0	(23)
(24)			MCWB	11.2	12.2	13.7	14.8	16.1	18.3	19.1	18.7	17.7	15.2	13.0	11.6	(24)
(25)		10%	DB	14.2	16.1	19.0	21.4	24.7	29.3	31.2	30.6	30.2	25.4	18.4	14.6	(25)
(26)			MCWB	10.5	11.2	12.6	13.6	15.1	17.1	18.5	18.1	16.9	14.5	12.1	10.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.4	17.2	18.6	20.1	21.6	22.6	21.8	21.6	18.9	16.6	15.3	(27)
(28)			MCDWB	16.7	19.6	24.6	27.7	32.9	36.1	37.3	35.9	32.0	29.5	22.7	17.4	(28)
(29)		2%	WB	13.0	14.1	15.6	16.9	18.3	20.1	20.7	20.2	19.6	17.5	15.4	14.0	(29)
(30)			MCDWB	15.5	18.4	22.1	26.0	29.6	33.6	33.8	33.2	32.0	27.3	20.6	16.0	(30)
(31)		5%	WB	12.0	13.0	14.5	15.3	16.8	18.9	19.7	19.2	18.6	16.5	14.4	12.8	(31)
(32)			MCDWB	14.1	16.9	20.1	23.0	26.5	31.1	32.1	30.9	30.5	25.2	18.6	14.7	(32)
(33)		10%	WB	11.0	11.9	13.2	14.0	15.5	17.7	18.8	18.4	17.7	15.5	13.4	11.5	(33)
(34)			MCDWB	13.6	15.4	17.9	20.4	23.4	28.2	30.3	29.3	28.6	22.9	17.2	13.5	(34)
(35)	Mean Daily Temperature Range		MDBR	10.1	10.7	11.9	12.5	13.3	15.2	16.2	16.0	15.8	14.4	11.9	10.2	(35)
(36)		5% DB	MCDBR	12.8	14.2	16.3	18.3	19.5	20.6	20.1	20.4	20.9	19.0	15.0	12.2	(36)
(37)			MCWBR	7.7	8.3	8.8	8.8	8.3	8.1	7.3	7.4	8.1	8.2	7.7	7.4	(37)
(38)			5% WB	MCDBR	9.3	11.7	14.7	16.3	17.9	19.1	18.7	18.3	18.1	15.9	12.2	8.0
(39)		MCWBR		6.3	7.3	8.5	8.3	8.1	7.8	7.3	7.0	7.3	7.0	5.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.325	0.339	0.359	0.365	0.360	0.365	0.350	0.350	0.333	0.325	0.324	(40)	
(41)		taud	2.527	2.518	2.459	2.383	2.388	2.410	2.398	2.431	2.426	2.507	2.510	2.520	(41)	
(42)		Ebn at Noon	867	910	925	920	915	916	908	917	900	886	852	835	(42)	
(43)		Edh at Noon	78	89	103	118	119	117	117	111	105	88	78	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.36	3.34	4.74	6.18	7.29	8.06	7.82	7.01	5.81	4.22	2.84	2.10	(44)	
(45)		RadStd	0.34	0.40	0.44	0.52	0.43	0.31	0.22	0.23	0.20	0.26	0.17	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.55	N/A	N/A	N/A	N/A	N/A	-28	-130	+198	+60	(46)
(47)	Regional (44 neighbors)	N/A	N/A	-1.17	N/A	N/A	N/A	N/A	-110	+135	N/A	(47)

Nomenclature: See separate page